R Worksheet: Matrices, Plots, Libraries, and Functions

Section 1: Matrices + Operations

Three students take a class together and study for a **project**, **midterm**, and **final** together.

- Student A scores: `12, 23, 34`
- Student B scores: `45, 56, 67`
- Student C scores: `78, 89, 64`

Q1: Create a matrix containing their scores.
Use `matrix()` to create the 3x3 matrix:

```
# Your code here:
$
$
$
$
$
$
$
$
```

Q2: Name the rows and columns
Use `colnames()` and `rownames()` to label rows and columns.

```
# Your code here:
$
$
```

Q3: Compute each student's average score
Which student has the highest average?

```
# Your code here:
$
```

Plots

Q4: How can we make simple graphs in Base R?
What functions can we use to create histograms, scatterplots, or boxplots?

```

```

Q5: Create a histogram using the matrix scores
Hint: Convert the matrix to a vector first using `as.vector()`.

```
# Your code here:
$
```

Q6: Use the built-in `airquality` dataset

This dataset contains daily New York air quality measurements from May–September 1973.

- `Ozone`: Ozone concentration (ppb)
- `Solar.R`: Solar radiation (langley)
- `Wind`: Wind speed (mph)
- `Temp`: Temperature (°F)
- `Month`: Month (5 = May, …, 9 = September)
- `Day`: Day of the month

a) Histogram of Ozone

```
# Your code here:
$
```

b) Scatterplot: Ozone vs Temperature

```
# Your code here:
$
$
$
$
```

c) Boxplot: Temperature by Month

```
# Your code here:
$
$
$
$
```

Intro to Libraries

Q7: Install and load the libraries

Install and load the `igraph` and `networkD3` packages.

```
# Your code here:
$
$
$
$
```

Q8: What are `igraph` and `networkD3`?

How are these libraries useful for network analysis and visualization?

Write code to access each package's documentation.

```
# Your code here:
$
$
```

Q9: Explore usage

Run these examples and describe what they do.

a) `igraph` example

```
g <- graph(edges = c(1,2, 2,3, 3,1), n = 3, directed = FALSE)
plot(g)
```

b) `networkD3` example

```
simpleNetwork(data.frame(source = c("A", "B", "C"),
                         target = c("B", "C", "A")))
```

Functions

Q10: Write a function to simulate rolling a die without replacement

```
# Your code here:
$
$
$
$
```